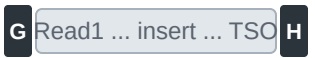
a Mechanism (schematic)

Each position PCR puts the lower-index adapter on the Read1 end and the higher-index adapter on the TSO end:



a proper molecule of slot G to H; adapters join only to the same letter, so arrays grow A, B, ..., H, Q

A TSO-TSO molecule is primed by the TSO primer at both ends and gets the higher-index adapter twice:

each H end then takes a slot H to Q molecule, so both halves run to Q

The read: a mirror junction (H' followed by H) with the TSO-TSO molecule at its centre